Gouki Minegishi

Ph.D. Candidate · The University of Tokyo

Tokyo, Japan · minegishi@weblab.t.u-tokyo.ac.jp · gouki510.github.io

RESEARCH INTERESTS

Mechanistic Interpretability · In-Context Learning · Large Language Models · Reasoning · Sparse Autoencoders · Neural Network Circuits

EDUCATION

Ph.D. in Computer Science — Apr 2025 – Mar 2028
The University of Tokyo · Advisor: Prof. Yutaka Matsuo

M.S. in Computer Science — Apr 2023 – Mar 2025
The University of Tokyo

B.S. in Engineering — Apr 2019 – Mar 2023
The University of Tokyo

SELECTED PUBLICATIONS

[1] **Mechanism of Task-oriented Information Removal in In-context Learning**
Hakaze Cho, Haolin Yang, **Gouki Minegishi**, and Naoya Inoue
International Conference on Learning Representations (ICLR), 2026

[2] **RL Squeezes, SFT Expands: A Comparative Study of Reasoning LLMs**
Kohsei Matsutani, Shota Takashiro, **Gouki Minegishi**, Takeshi Kojima, Yusuke Iwasawa, and Yutaka Matsuo
International Conference on Learning Representations (ICLR), 2026

[3] **Topology of Reasoning: Understanding Large Reasoning Models through Reasoning Graph Properties**
Gouki Minegishi, Hiroki Furuta, Takeshi Kojima, Yusuke Iwasawa, and Yutaka Matsuo
Neural Information Processing Systems (NeurIPS), 2025

[4] **Beyond Induction Heads: In-Context Meta Learning Induces Multi-Phase Circuit Emergence**
Gouki Minegishi, Hiroki Furuta, Shohei Taniguchi, Yusuke Iwasawa, and Yutaka Matsuo
International Conference on Machine Learning (ICML), 2025

[5] **Rethinking Evaluation of Sparse Autoencoders through the Representation of Polysemous Words**
Gouki Minegishi, Hiroki Furuta, Yusuke Iwasawa, and Yutaka Matsuo
International Conference on Learning Representations (ICLR), 2025

[6] **Interpreting Multi-Attribute Confounding through Numerical Attributes in Large Language Models**
Hirohane Takagi*, **Gouki Minegishi***, Shota Kizawa, Issey Sukeda, Hitomi Yanaka (* Equal Contribution)
International Joint Conference on NLP / Asia-Pacific Chapter of ACL (IJCNLP-AACL), 2025

[7] **Bridging Lottery Ticket and Grokking: Understanding Grokking from Inner Structure of Networks**
Gouki Minegishi, Yusuke Iwasawa, and Yutaka Matsuo
Transactions on Machine Learning Research (TMLR)

[8] **Towards Empirical Interpretation of Internal Circuits and Properties in Grokked Transformers on Modular Polynomials**
Hiroki Furuta, **Gouki Minegishi**, Yusuke Iwasawa, and Yutaka Matsuo
Transactions on Machine Learning Research (TMLR)

[9] **ADOPT: Modified Adam Can Converge with Any β2 with the Optimal Rate**
Shohei Taniguchi, Keno Harada, **Gouki Minegishi**, et al.
Neural Information Processing Systems (NeurIPS), 2024

RESEARCH & WORK EXPERIENCE

Research Scientist Jun 2025 – Present

Third Intelligence, Inc.

Research on developing new forms of intelligence beyond conventional AI, aiming to bridge human-like reasoning, perception, and action.

Chief AI Engineer Nov 2024 – May 2025

Matsuo Institute, Inc.

Developed automated driving software, including 3D object detection with deep learning and integration of the World Model into automated driving.

Research Intern Aug 2024 – Oct 2024

Preferred Networks, Inc.

Research on in-context learning in foundation models, with knowledge conflicts.

ACADEMIC HONORS

Young Researcher Encouragement Award, NLP 2025

Outstanding Presentation Award, JSAI 2024 2024

Oral Presentation, ICLR 2025 Workshop (Building Trust in LMs) 2025

Oral Presentation, ICLR 2024 Workshop (Bridging Practice and Theory) 2024

Selected for "BOOST NAIS" Scholarship Program for fostering advanced AI talents

ACADEMIC SERVICE

Reviewer: ICLR 2026, AAAI 2026, NeurIPS 2025, ICML 2025/2026

Organizer: Mechanistic Interpretability Organized Session at JSAI 2025